Wenjia Ge

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Education

Shenzhen University *09/2020 - 07/2024*
Bachelor of Science in Physics Shenzhen, China

- GPA: 3.65/4.00

New York University *09/2024 - Expected 12/2026*
Master of Science in Electrical and Electronics Engineering New York, United States of America

- GPA: 3.926/4.00
- **Relevant Coursework:** Digital Signal Processing, Probability and Stochastic Processes, Real Time Embedded Systems, Computing Systems Architecture, Electromagnetic Theory and Applications, Applied Matrix Theory

Work Experience

AI^2 Robotics (Shenzhen) Science and Technology Co., Ltd. *10/2025 - 01/2026*
Research Intern, Embodied AI and Robotics Shenzhen, China

- Supported research and engineering work for embodied AI and general purpose service robots, including system testing, technical research, and performance review.
- Conducted subsystem testing and performance analysis to evaluate robot behavior, identify failure cases, and support validation and optimization workflows.
- Collaborated with algorithm, hardware, and systems engineers to review technical requirements, communicate test findings, and improve system reliability.
- Researched industry benchmarks and competitor products, prepared technical materials, and supported partner visits and robotics product discussions.

Shenzhen Zhi'ao Technology Co., Ltd. *07/2023 - 08/2023*
Hardware Test Engineer Intern, Technical Department Shenzhen, China

- Conducted functional and performance testing on new hardware products to evaluate reliability, identify defects, and verify compliance with technical specifications.
- Used automated testing tools and basic scripting to improve test efficiency, expand test coverage, and support repeatable validation workflows.
- Collaborated with hardware and software engineers to troubleshoot product issues, summarize test results, and recommend improvements that reduced hardware failure rates by 15%.

Bank of Communications, Shenzhen Branch *01/2023 - 03/2023*
Intern, Finance Department Shenzhen, China

- Supported daily accounting operations and financial report preparation, ensuring data accuracy and compliance with internal procedures.
- Assisted with financial analysis, budget tracking, and cost review, using financial software to organize data and prepare summary materials.

PwC Zhong Tian LLP *07/2022 - 08/2022*
Intern, Accounting Department Shenzhen, China

- Assisted with financial statement review and audit procedures by checking client records for completeness, consistency, and regulatory compliance.
- Processed and analyzed client financial data, prepared audit documentation, and communicated with team members to support efficient project completion.

Projects

Point Cloud Data Processing for Robot Training *2026 – Present*
Ongoing Research Project, AI4CE Lab, New York University New York, NY

- Collecting visual and spatial data using MetaCam to support robot perception and training workflows.
- Processing raw sensor data to generate spatial point cloud representations for robotic scene understanding.
- Building data preprocessing and visualization workflows to improve the structure, quality, and usability of robotic training data.

Sensor Based Medical Robotics Simulation *2025 – Present*
Research Project New York, NY

- Developed a medical robotics simulation workflow using 3D Slicer, VMTK, ROS2, RViz, and SlicerROS2 for patient specific vascular modeling and robot visualization.
- Processed CTA based anatomical models and generated vessel centerlines using landmark selection to support path planning and pre surgical simulation.
- Integrated sensor and device data into the ROS2 pipeline to support real time visualization and system validation.
- Analyzed tactile and contact sensor signals to improve interpretation of robot interaction with biological environments.

Order of Accuracy of Spatial Discretization of Method of Characteristics *11/2022 – 09/2023*
Undergraduate Research Project Shenzhen, China

- Analyzed the spatial discretization accuracy of the Method of Characteristics, focusing on Flat Source and Linear Source approximations.
- Implemented the Method of Manufactured Solutions to validate numerical accuracy in planar geometry.
- Conducted data collection and error analysis to evaluate accuracy trends across spatial meshes and characteristic rays.
- Published the results as first author in the proceedings of the 30th International Conference on Nuclear Engineering.

Experimental Study of Semiconductor Thin Films *12/2023 – 06/2024*
Undergraduate Thesis Project Shenzhen, China

- Conducted fabrication, post treatment, and characterization of semiconductor thin films to evaluate crystallinity, surface properties, and structure property relationships.
- Improved thin film quality through interface engineering, reducing surface roughness by approximately 18%, and received the Outstanding Undergraduate Thesis Award.

Publications

- Ge Wenjia. (2023). First Author. **Order of Accuracy of Spatial Discretization of Method of Characteristics**. *Published in 30th International Conference on Nuclear Engineering (ICONE30), May 21-26, 2023, Kyoto, Japan.*

Skills

- **Programming:** Python, C, MATLAB
- **Robotics and Embedded Systems:** ROS2, RViz, 3D Slicer, SlicerROS2, VMTK, Arduino, PlatformIO, Simulink, MetaCam
- **Data and Research:** Point cloud processing, sensor signal processing, numerical analysis, scientific computing, data analysis, lab instrumentation
- **Languages:** Chinese native, English fluent
- **Certifications and Tests:** ACCA F1, F2, F3, TOEFL 107, GRE 327